

The Corporate Blog (TCB)

Project Architecture & Execution Plan

Version 1.0 · Production-Grade · SEO-First · Target: 1M+ Daily Active Users

Reference App & Test Credentials

Live URL: <https://blueblog-five.vercel.app/>

Role	Email	Password
Admin	admin@gmail.com	admin@123
Writer	shraddha@gmail.com	Pass@123
Editor	yash@gmail.com	Pass@123

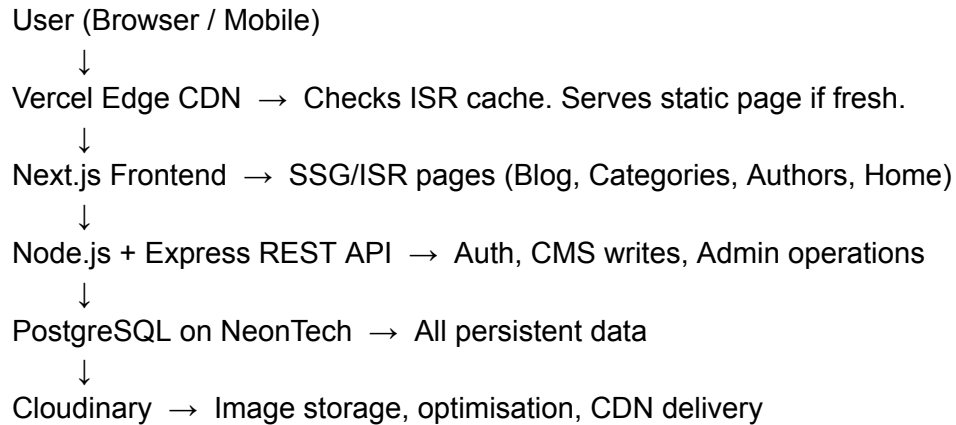
1. Core Goals (In Priority Order)

These goals drive every technical decision. When trade-offs arise, refer to this order.

1. **SEO First** — Every page must be indexable and rankable. No SEO = no organic traffic. This is non-negotiable.
 2. **Performance** — Core Web Vitals directly affect Google rankings. Target Lighthouse 90+ on all metrics.
 3. **Scalability** — Architecture must support 1M+ Daily Active Users from day one. Retrofitting is expensive.
 4. **Clean Authoring Experience** — Writers and editors need an intuitive CMS so they can focus on content, not the tool.
 5. **Monetization-Ready** — Ad slots, affiliate hooks, and premium content must be easy to add later.
 6. **Cost Control** — Startup budget. Use serverless infrastructure that scales to zero when idle.
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2. High-Level Architecture

Here's how a user request flows through the system:



Key principle: Public pages are always served from CDN cache. The API is only called for admin, auth, and write operations — keeping costs low and performance high.

3. Tech Stack

Layer	Technology	Why
Frontend	Next.js (App Router)	SSG, ISR, file-based routing, built-in SEO support
Styling	Tailwind CSS	Utility-first, no runtime overhead, fast to build
Backend	Node.js LTS + Express	Stable, widely understood, stateless REST API
Database	PostgreSQL on NeonTech	Serverless, auto-scales, generous free tier
ORM	Prisma + PGBouncer	Type-safe queries, connection pooling
Auth	JWT + Refresh Tokens	Stateless, scalable, no session storage needed
Writer Auth	Google OAuth via Auth0	Self-registration without admin involvement
Images	Cloudinary	Auto WebP/AVIF, lazy load, responsive sizes, CDN

Hosting	Vercel	Git push deploys, built-in ISR, edge network
Cache	Vercel ISR → Redis (Phase 2)	ISR for pages, Redis for hot posts later
SEO Tools	Next SEO + Custom Meta Engine	Dynamic meta tags, OpenGraph, Schema.org
DNS / CDN	Cloudflare	DDoS protection, free tier covers most needs
Monitoring	Sentry (Phase 2)	Runtime error tracking

4. SEO Strategy (Non-Negotiable)

SEO is the primary growth channel. Every frontend decision must preserve or improve it.

4.1 Rendering Strategy Per Page Type

Page	Strategy	Reason
Blog Home	SSG + ISR (15 min)	Fast load, always fresh for crawlers
Category Pages	SSG + ISR (15 min)	High-value ranking pages
Blog Detail Pages	SSG + ISR (15 min)	Core SEO pages — edits reflect within 15 min
Author Pages	SSG	Author authority pages — rarely change
Admin / CMS	CSR only	Private, never indexed — no SSR needed

ISR Cache Rule:

- Existing posts → Revalidated every 15 minutes
- New posts → Appear instantly (on-demand revalidation triggered on publish)
- Drafts → Never exposed in sitemap or to crawlers

4.2 Required SEO Elements on Every Page

Element	How It's Implemented
<code><title></code> tag	Dynamic per page, auto-generated from post title
<code><meta description></code>	Manual field in CMS; auto-fallback from excerpt

<code><link rel="canonical"></code>	Always set explicitly — prevents duplicate content penalties
OpenGraph + Twitter Cards	og:title, og:image, og:description — required for social previews
Schema.org Article	Structured data for Google News and rich search results
Schema.org Breadcrumb	BreadcrumbList — shows breadcrumbs in Google SERP
Schema.org Author	Author authority for E-E-A-T ranking signals
XML Sitemap (dynamic)	Auto-generated server-side — only PUBLISHED posts included
robots.txt	Blocks admin/draft URLs, allows all public content
Clean Slug URLs	<code>/blog/post-title · /blog/category/slug · /blog/author/slug</code>

5. Database Design

PostgreSQL on NeonTech. All tables use UUID primary keys.

5.1 Users Table

Stores all platform users (Admins, Editors, Writers).

Column	Type	Notes
id	UUID	Primary Key
name	VARCHAR	Display name
email	VARCHAR UNIQUE	Login email
role	ENUM	ADMIN, EDITOR, or WRITER
bio	TEXT	Used on author SEO pages
profile_image	TEXT	URL to avatar
created_at	TIMESTAMP	—

5.2 Posts Table

Core content table. Slug and author_id are indexed for fast lookups.

Column	Type	Notes
id	UUID	Primary Key
title	VARCHAR	Post headline
slug	VARCHAR UNIQUE	URL slug — indexed
excerpt	TEXT	Summary; used as meta description fallback
content	JSONB	Structured block content (NOT raw HTML)
banner_image_id	UUID FK	References images table
author_id	UUID FK	References users table — indexed
status	ENUM	DRAFT or PUBLISHED
seo_title	VARCHAR	Overrides title in <code><title></code> tag
seo_description	TEXT	Overrides excerpt in meta description
canonical_url	VARCHAR	Override if cross-posted elsewhere
published_at	TIMESTAMP	Indexed for date sorting
created_at	TIMESTAMP	—

5.3 Categories Table

Column	Type	Notes
id	UUID	Primary Key
name	VARCHAR	e.g. "Software Engineering"
slug	VARCHAR UNIQUE	e.g. "software-engineering"

5.4 Post-Category Mapping

Many-to-many join. One post can belong to multiple categories.

Column	Type
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post_id UUID FK → posts

category_id UUID FK → categories

5.5 Images Table

Central image registry with full SEO metadata per image.

Column	Type	Notes
id	UUID	Primary Key
url	TEXT	Cloudinary URL
alt_text	TEXT	★★★★★ Most important SEO field
title	TEXT	Image title attribute
caption	TEXT	Visible caption — crawled by Google
width / height	INT	Required to prevent layout shift (CLS)
created_at	TIMESTAMP	—

6. Roles & Permissions

Permission	Admin	Editor	Writer
Manage users	✓	✗	✗
Publish / Unpublish posts	✓	✓	✗
SEO field overrides	✓	✓	✗
Category management	✓	✗	✗
Approve / Reject posts	✓	✓	✗
Schedule publishing	✓	✓	✗
Create & edit drafts	✓	✓	✓
Upload images	✓	✓	✓

Preview SEO score	✓	✓	✓
Self-register via Google	—	—	✓ (Auth0)

7. Content Editor Capabilities

Content is stored as **structured JSON blocks**, not raw HTML. This makes it safe, portable, and extensible.

7.1 Supported Block Types

Block	Notes
Headings H1–H6	Used to auto-generate Table of Contents
Bold / Italic / Underline	Inline text formatting
Ordered & Unordered Lists	SEO-friendly content structure
Tables	Very SEO-friendly — Google reads table data well
Inline Images	With full SEO metadata (see 7.2 below)
YouTube Embed	Lazy-loaded iframe — no performance penalty
Blockquotes	Styled pull quotes
Code Blocks	Syntax-highlighted (Phase 2)

7.2 Image SEO Fields (Critical)

Every image uploaded must fill these fields in the CMS:

Field	SEO Priority	Purpose
alt_text	★★★★★ Critical	Google Image Search + screen readers
image_url	★★★★★ Required	Cloudinary URL with auto WebP/AVIF
title	★★★☆☆ High	Tooltip text + keyword signal
caption	★★★☆☆ High	Visible text crawled as page content

width & height	★★★★☆ CLS	Prevents layout shift — Core Web Vitals
format (WebP/AVIF)	★★★★☆ Speed	30–50% smaller than JPEG
lazy_load	★★★★☆ LCP	Defers off-screen images — speeds up page

8. Backend API Design

Express REST API. All inputs validated with Zod. All routes protected by role middleware.

8.1 Key Endpoints

Endpoint	Access	Description
POST /auth/login	Public	Email + password → JWT + refresh token
POST /auth/google	Public	Google OAuth for Writer self-registration
POST /posts	Writer+	Create a new draft post
PUT /posts/:id	Writer+	Update post content or metadata
PUT /posts/:id/publish	Editor+	Set status to PUBLISHED — triggers ISR
GET /posts/slug/:slug	Public	Fetch post by slug — used by frontend SSG/ISR
GET /categories/:slug/posts	Public	Published posts in a category (paginated)
GET /authors/:slug/posts	Public	All posts by a specific author
GET /sitemap.xml	Public	Dynamic sitemap — only PUBLISHED posts
GET /admin/users	Admin only	List and manage all users

8.2 API Standards

- **Zod validation** — Every request body is validated before hitting the database
- **Central error handler** — Consistent error format across all endpoints
- **Rate limiting** — Applied to public and auth endpoints to prevent abuse
- **Pagination** — All list endpoints support `page` and `limit` query params
- **Role middleware** — ADMIN > EDITOR > WRITER hierarchy enforced per route

- **Stateless** — No server sessions; JWT verified on every request
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9. Performance & Scalability

Target: **Lighthouse 90+ on Performance, SEO, Accessibility, and Best Practices.**

9.1 Frontend

- **Vercel ISR** — 99% of requests served from CDN; near-zero server load
- **Cloudinary** — Auto-resize, WebP/AVIF conversion, lazy loading
- **Brotli compression** — Enabled by default on Vercel
- **Critical CSS inlined** — Above-the-fold styles load instantly
- **next/font** — Zero layout shift on custom web fonts
- **Shimmer skeleton screens** — Shown while loading; gives native app feel on mobile

9.2 Backend

- **Stateless APIs** — Scale horizontally on Vercel Serverless with zero config
- **NeonTech connection pooling** — Handles DB connection bursts automatically
- **Redis caching** (Phase 2) — Hot posts and trending content cached in memory

9.3 Database Indexes

Index	Reason
posts.slug	Primary lookup for blog detail pages
posts.published_at	Sorting recent and featured posts
posts.author_id	Author page queries
posts.content (JSONB GIN)	Future full-text content search
categories.slug	Category page queries

10. Advanced SEO Features (Phase 2)

Feature	Benefit
Internal Linking Engine	Auto-suggest related posts during writing to improve link graph

Related Posts Widget	Based on shared categories — reduces bounce rate
Popular Posts (view counter)	Drives recirculation traffic to trending content
Breadcrumbs with Schema	BreadcrumbList structured data — improves SERP CTR
FAQ Schema Blocks	Writers add FAQ sections — eligible for Google FAQ rich results
Auto Table of Contents	Generated from H2/H3 headings — improves time-on-page

11. Deployment & Infrastructure

Component	Platform	Notes
Frontend	Vercel	Git push auto-deploys, ISR built-in
Backend API	Vercel Serverless	Scales to zero, no ops overhead
Database	NeonTech	Serverless PostgreSQL, auto-pause
Images	Cloudinary	Auto-optimises, generous free tier
DNS & Security	Cloudflare	DDoS protection, free plan covers early stage

Cost note: The entire stack runs near-zero cost at low traffic. NeonTech and Cloudinary free tiers cover early-stage usage. Upgrade Vercel hobby → pro only when traffic demands it.

12. Monitoring & Quality

Tool	Purpose
Lighthouse CI	Run on every PR — block merge if score drops below 90
Google Search Console	Monitor indexing, clicks, impressions, Core Web Vitals
CrUX (Core Web Vitals)	Real user LCP, FID, CLS data from Google

Sentry (Phase 2)	Runtime error tracking — JS errors + API 5xx alerts
Vercel Access Logs	Track slow routes and error rate spikes

13. Additional Requirements Summary

Requirement	Implementation
Shimmer loading effect	Skeleton screens on all dynamic content — no blank flash
Native mobile feel	Touch-optimised, no horizontal scroll, fast tap response
Dynamic sitemap	Server-generated — drafts are NEVER included or exposed
Google Auth (Writers)	Writers self-register via Google OAuth — no manual setup
Lighthouse 90+	Enforced on every PR via Lighthouse CI
15-min ISR cache	Edits appear within 15 min. New posts appear instantly.
Modern, clean UI	Reusable component design system — consistent across all pages
